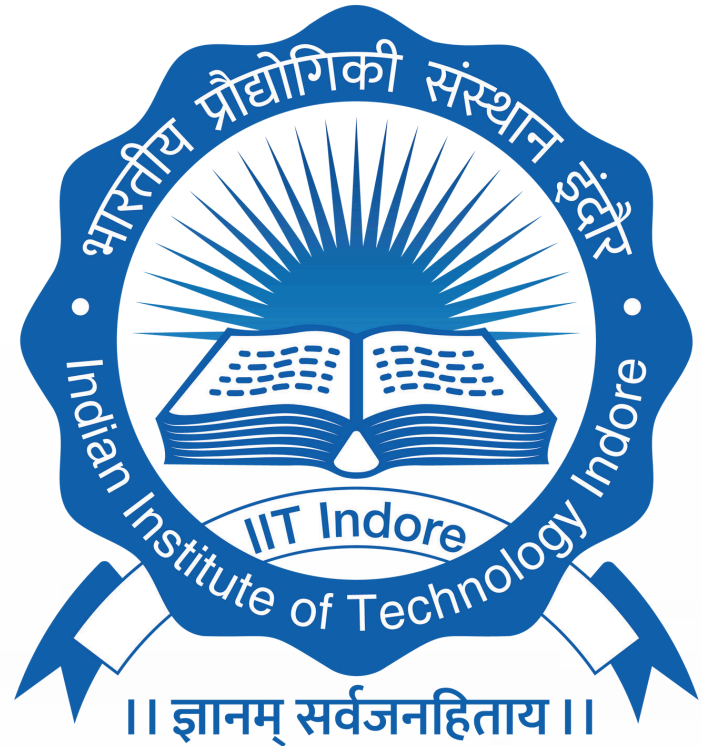




Constraining Reionization Parameters from 21-cm Power Spectrum through BNN-based Emulator using Bayesian Inference

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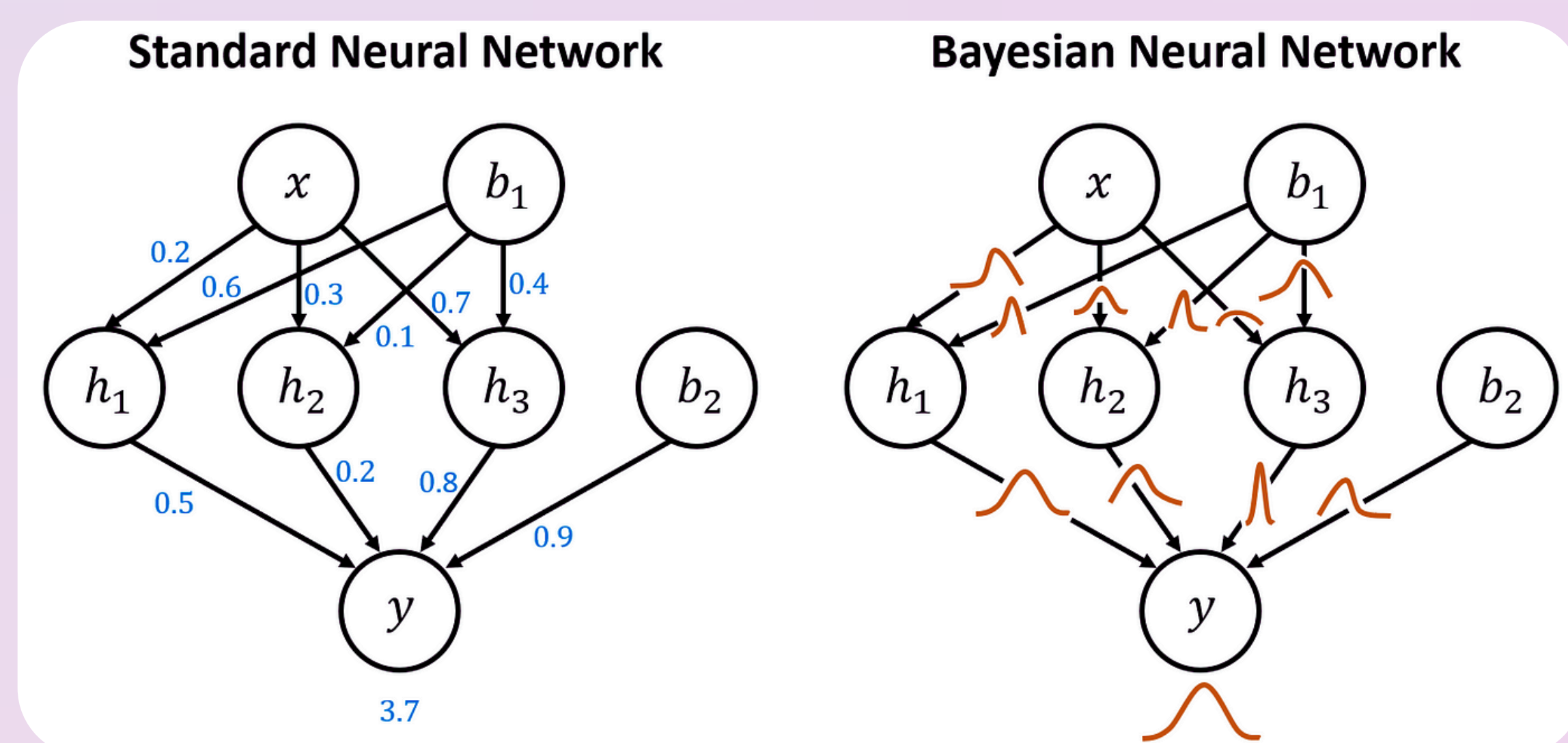
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1. Motivation

The Epoch of Reionization (EoR) is one of the least understood periods in the universe's history, yet understanding it is crucial for uncovering the origins of cosmic structure. Observations of the redshifted 21-cm signal by next-generation radio telescopes like the Square Kilometre Array (SKA) hold the promise of shedding light on this epoch. However, the interpretation of these signals is computationally intensive due to the complexity of reionization models. Current methods, such as Artificial Neural Networks (ANNs), struggle with bias and lack uncertainty estimates, limiting their reliability for parameter inference. This motivates our development of advanced Neural Network-based emulators, including Bayesian Neural Networks (BNNs), to enhance accuracy, provide robust uncertainty quantification, and enable reliable interpretation of SKA-like observations in realistic scenarios.

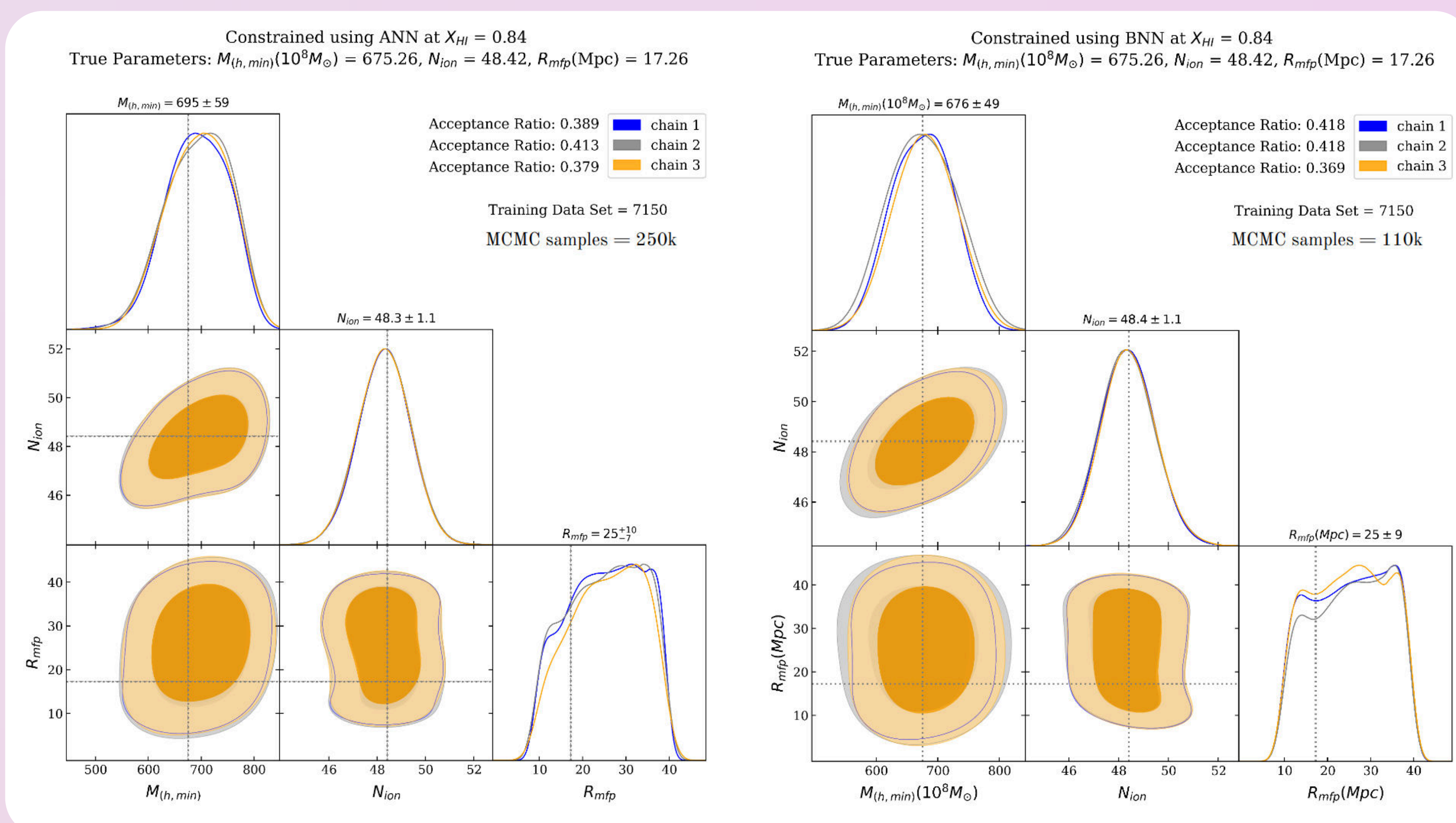
3. ANN & BNN

Artificial Neural Networks (ANNs) learn weights and biases as fixed point values, leading to single-point predictions without uncertainty. In contrast, Bayesian Neural Networks (BNNs) learn these parameters as posterior distributions, enabling predictions as distributions. This allows for uncertainty quantification, making BNNs more robust and suitable for inference.

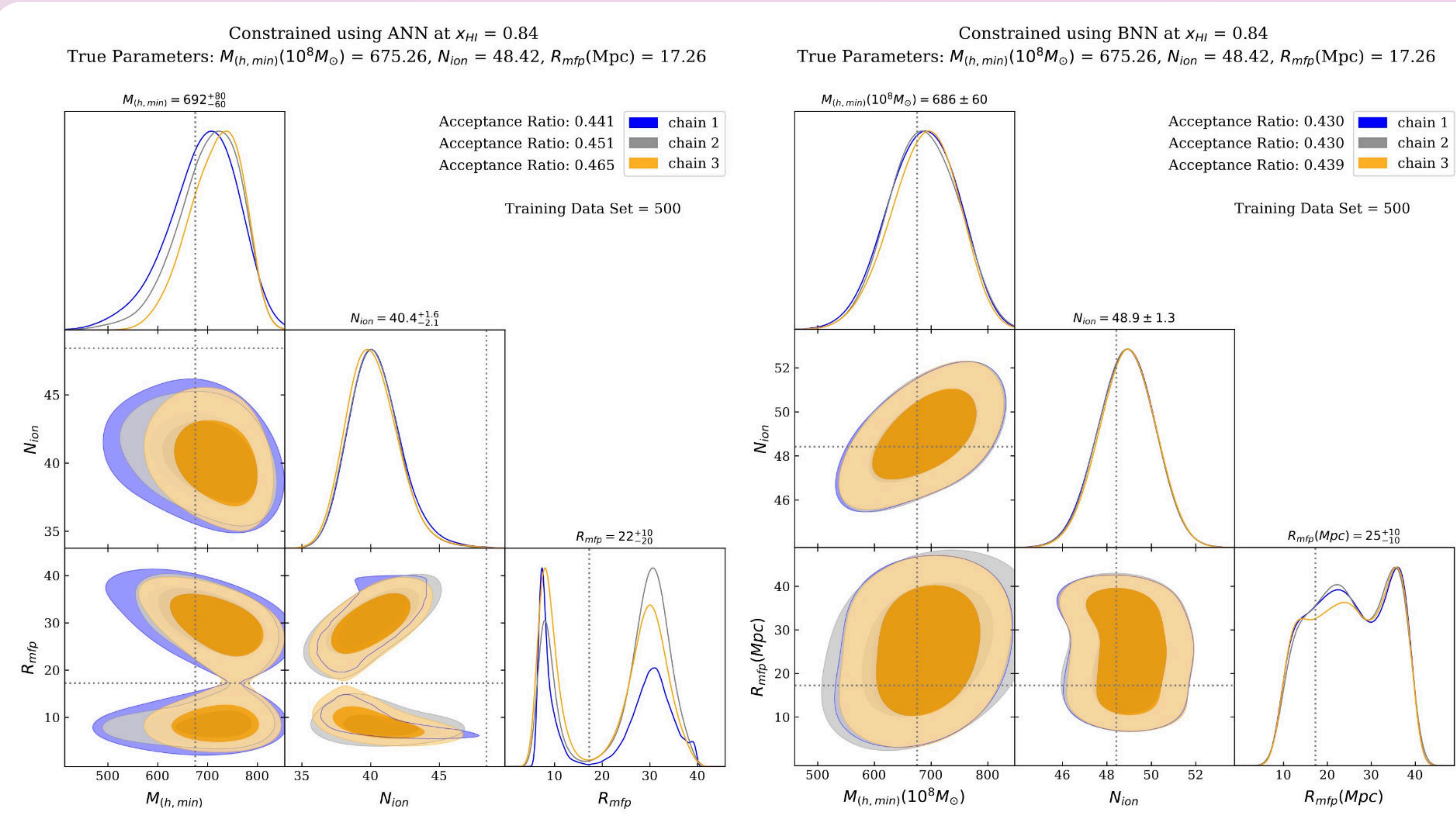


5. Constrained Parameters

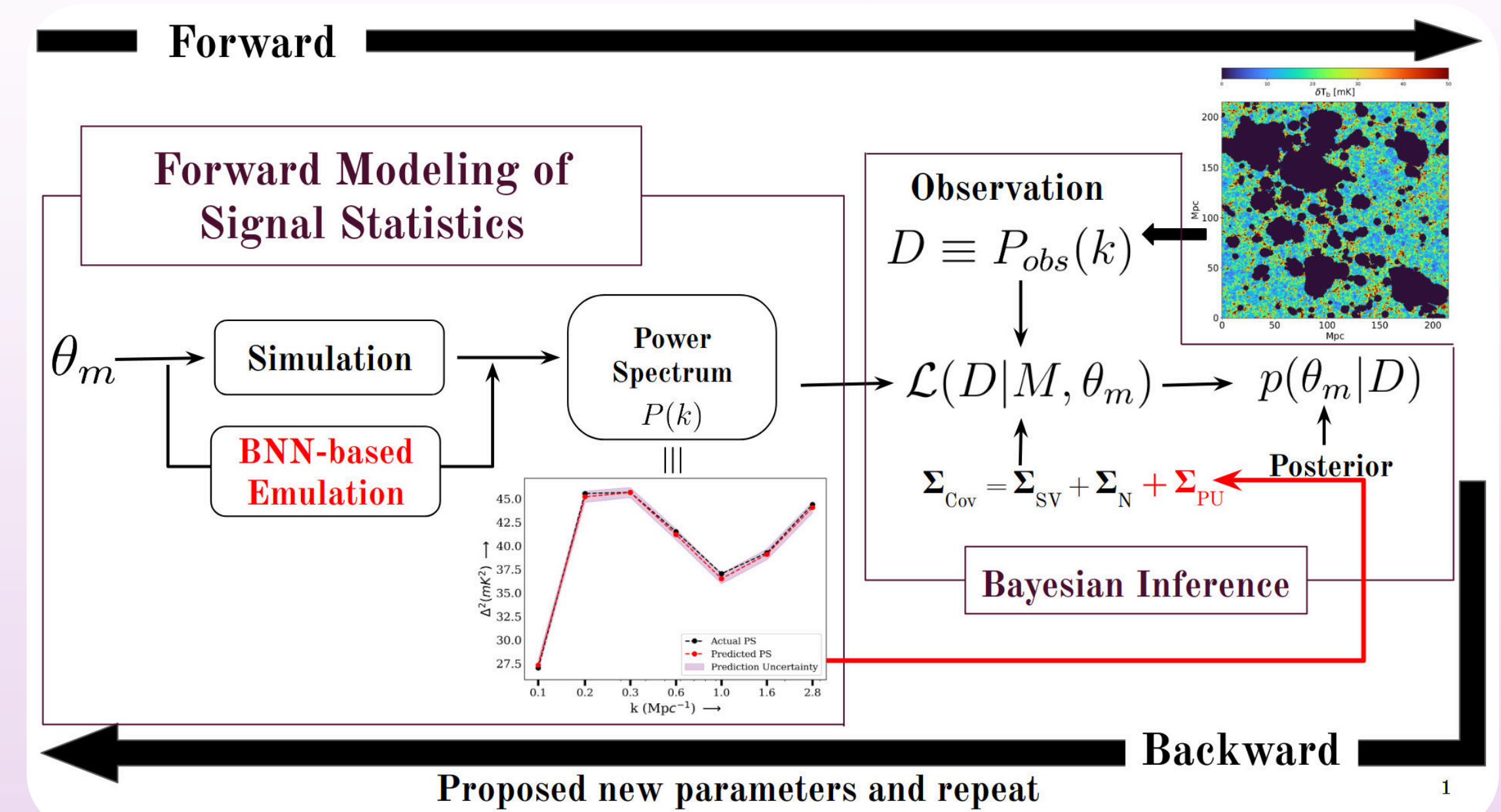
a) Large Training Dataset



b) Small Training Dataset



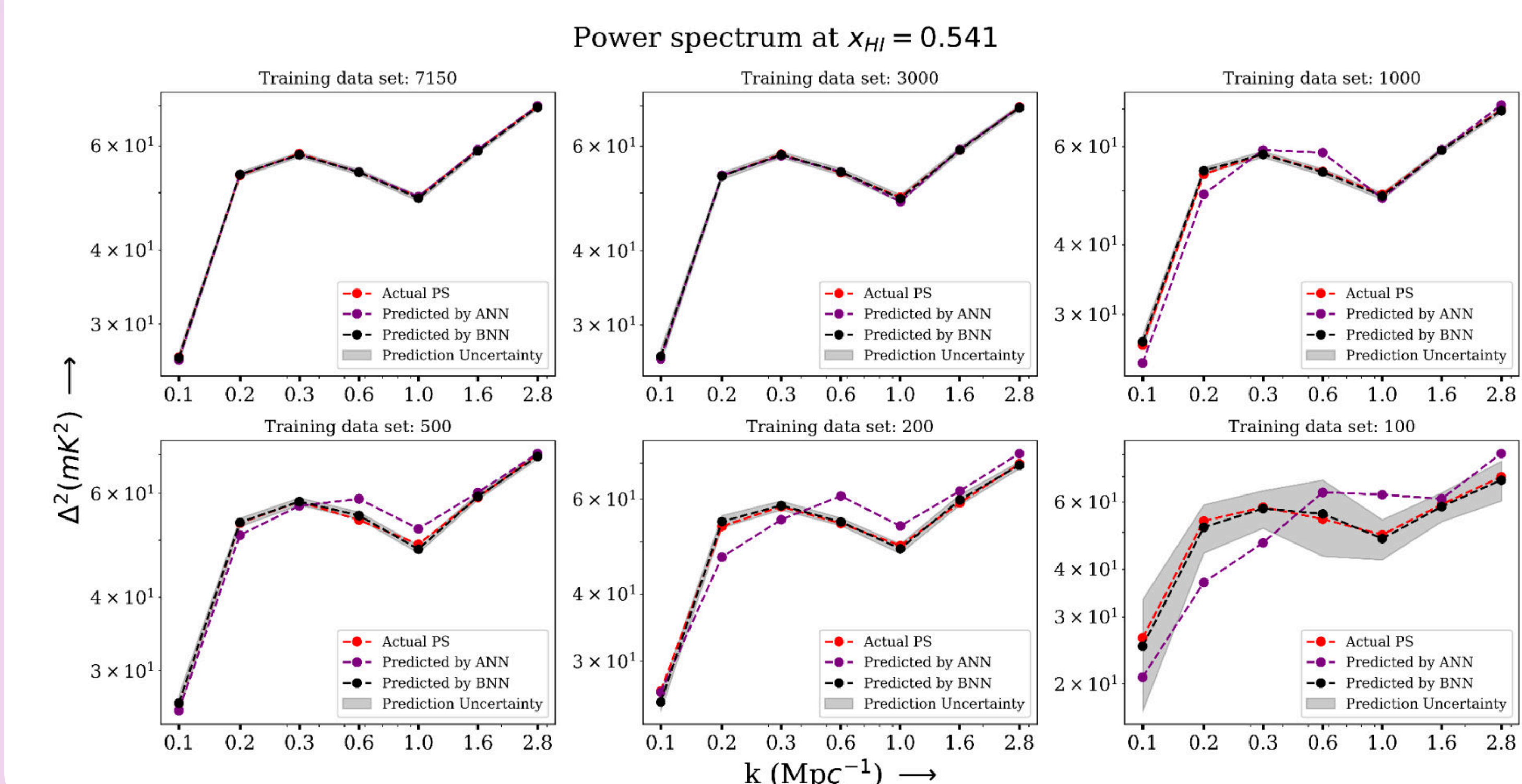
2. Inference Pipeline



4. Power Spectrum Predictions

During training, ANNs optimize their parameters to best fit the data, while BNNs incorporate Bayesian inference to learn data trends. With large training datasets, both perform similarly (though BNNs also provide uncertainty). However, as the dataset size decreases, ANNs become biased toward the training data, leading to incorrect predictions. In contrast, BNNs remain accurate within their uncertainty range, even as the uncertainty increases with smaller datasets.

Comparison of ANN & BNN-emulator prediction with different training data sets sizes



learn more

6. Conclusions

- BNN-based emulators can quantify uncertainty in their predictions, which is incorporated during inference.
- Unlike ANNs, which become biased with smaller datasets, BNNs remain robust and accurate.
- BNNs provide better parameter constraints than ANN-based emulators by incorporating uncertainty into the Bayesian Inference (MH-MCMC) pipeline.
- BNNs are particularly useful in scenarios with limited observational data, as they can effectively constrain parameters even with small datasets.
- BNNs require fewer MCMC samples than ANNs to achieve convergence when inferring model parameters, improving computational efficiency.